



Semester 2 Examinations 2016-17

Exam Codes	3BA1, 1OA1, 4BCT1, 4BMS2, BME1, 4BS2
Exams	BSc Arts, Occasional Arts, BSc Science & Mathematical Science BSc Computer Science & Technology
Module	Cryptography
Module Code	CS402, MA492
Paper No	1
Repeat Paper	No
External Examiner	Prof. M. Lawson
Internal Examiner	Prof. G. Ellis Dr D. Degrijse*
Instructions	Answer all the questions. Each question carries the same number of marks.
Duration	2 hours
No. of Pages	2 pages (including this cover page)
School	Mathematics, Statistics and Applied Mathematics
Requirements:	
Release to Library:	Yes
Other Materials:	No

1.

- (a) Explain the difference between symmetric key and public key cryptography.
- (b) Discuss the difference between a block cipher and a stream cipher.

2.

- (a) Consider the RSA cryptosystem with public key (N, e) and private key (p, q, d) . Here p and q are the chosen secret prime numbers, $N = pq$, e is the encryption exponent and d is the decryption exponent. Explain how knowing d can be used to efficiently factor N .
- (b) If factoring large integers into primes is hard, then is it not just as hard to find large prime numbers? Explain briefly how this problem is dealt with in practice.

3.

- (a) In general it is a difficult problem to find points on a given elliptic curve. A consequence of this is that encoding a message as a point on an elliptic curve is also a non-obvious problem. Explain how one can message encode messages to elements of a given elliptic curve over a finite field of prime order.
- (b) Describe the El Gamal elliptic curve cryptography system.

4.

- (a) Explain the concept of ‘forward secrecy’ in relation to key encryption using RSA. Also discuss Diffie-Hellman key exchange and how it provides a better solution.
- (b) Explain what a ‘man in the middle attack’ is in the context of Diffie-Hellman key exchange.